NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2022-2023

EE6402 – REAL-TIME DSP DESIGN AND APPLICATIONS

April / May 2023 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 4 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Consider two Q3.3 fixed point (2's complement) numbers a and b.

   (i) Addition of a and b gives $c = a + b$. In order to avoid overflow, how many bits are required to represent c? What fixed point format is required to represent c?

   (ii) Let $a = 0010.101$ and $b = 1110.100$. Show binary addition of a and b to obtain c. Further, express c in decimal.

   (iii) Multiplication of a and b gives $d = ab$. In order to avoid overflow, how many bits are required to represent d? What fixed point format is required to represent d?

(13 Marks)

(b) An input signal $x(n)$ is rounded with a quantization step size of 2^{-7}. This rounded input $[x(n)]$ is filtered by an FIR filter with filter length 16. After each multiplication in the filter, the multiplier output is truncated with a quantization step size of 2^{-11}. The output of the filter $y(n)$ is rounded with a quantization step size of 2^{-7}. Let $e_{total}(n)$ denote the total output noise due to all the above quantization steps. Find the mean of $e_{total}(n)$.

(7 Marks)

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2. (a) Let $e_1(n)$ be a zero-mean white noise source with auto-correlation $R_1(m) = \frac{1}{8}\delta(m)$. Let $e_2(n)$ be another uncorrelated noise source with mean $\eta_2 = -\frac{1}{5}$, variance $\sigma_2^2 = \frac{1}{4}$, auto-correlation $R_2(m) = \frac{1}{25} + \frac{1}{4}\delta(m)$, and power spectral density $S_2(e^{j\omega}) = \frac{1}{4} + \frac{2\pi}{25}\delta(\omega)$. Find the mean, the variance, the auto-correlation, and the power spectral density of $e_1(n) + e_2(n)$.

(10 Marks)

(b) Consider a band-pass signal of bandwidth 6 KHz centered around 33 KHz.

(i) Find the Nyquist sampling frequency (assuming low-pass sampling) for the above signal.

(ii) Find one example of a sampling frequency less than the above Nyquist sampling for the above signal. Sketch the spectrum of the sampled signal with your example sampling frequency to show there is no aliasing.

(iii) Find the minimum possible sampling frequency for the above signal.

(10 Marks)

3. (a) Explain, with the help of a sketch of the power spectral densities, how oversampling reduces the quantization error in the sigma-delta ADC. A sigma-delta ADC operating at the Nyquist sampling frequency of a given signal results in a SQNR of 4dB. If the sampling frequency is doubled, what will be the SQNR of this ADC? (Hint: Assume no noise shaping is used in the sigma-delta ADC.)

(8 Marks)

(b) A reservation table involving 2 stages and 3 clocks is shown in Figure 1. Find the forbidden set of initiations for this scheduling task. Also find the maximum achievable throughput for this scheduling task. If possible, find a scheduling strategy to achieve this throughput.

		time		
		0	1	2
stages	1	A		A
	2		A	A

Figure 1

(12 Marks)

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4. (a) A data flow graph is shown in Figure 2. All nodes are labelled by their computation time, and all edges are labelled by the number of delays, if any.

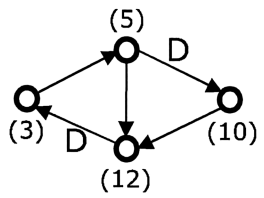

Figure 2

(i) Find the iteration bound of this graph.

(ii) Find the critical path of this graph.

(iii) Discuss if the feed-forward cutset based pipelining may be used to reduce the critical path of this graph.

(8 Marks)

(b) Consider the IIR filter $y(n) = x(n) + ay(n-2)$ shown in Figure 3, where the left node (add) takes T_A seconds, the bottom node (multiply by a) takes T_M seconds, and the right node (output Y) takes 0 second.

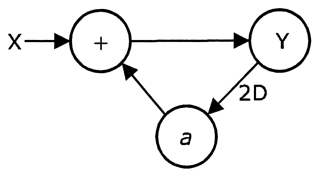

Figure 3

(i) What is the critical path, in seconds, of the graph in Figure 3?

(ii) Convert the graph to a parallel system with unfolding factor $J = 2$. Find the critical path.

(iii) Convert the graph to a parallel system with unfolding factor $J = 3$. Find the critical path. Is the critical path for $J = 3$ less than the critical path for $J = 2$? If not, why?

(12 Marks)

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5. (a) Bits 15-8 of *a* are available in the bits 7-0 of the auxiliary register **AR0**, while bits 7-0 of *a* are available in the bits 7-0 of the auxiliary register **AR1**. Suggest a way to use several TMS320C55x assembly language **MOV** instructions such that all 16 bits of *a* are available in any auxiliary register **ARn** in the correct order. You may use other registers and/or memory locations.

(10 Marks)

(b) Some auxiliary registers (AR) and accumulators (AC) in a TMS320C55x contain the following values:

```
XAR0      71_0002h
XAR1      00_8204h
AC0       33_3333_0000h
AC1       22_2222_0000h
```

Using the rules for moving bits between registers, find the value of the mentioned register after each of the following consecutive instructions:

MOV AR1, AR0	find the value of **XAR0**
MOV AR0, AC0	find the value of **AC0**
MOV XAR0, AC1	find the value of **AC1**

(10 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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